

# **L3a:** Equity Exchanges, Growth Rates, and Stylized Facts

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CHEME 5660 Cornell University

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**Risk.** Trading involves risk and is generally not recommended for those with limited resources, experience, or a low risk tolerance. Past performance does not guarantee future success.

**Responsibility.** You are responsible for your investment decisions based on your financial circumstances, objectives, risk tolerance, and liquidity needs.

# Objectives for Today

Today we connect equity-market infrastructure to price data, convert prices into growth rates and returns, and use recurring empirical patterns to test a model of equity behavior.

## Today's concepts

- **Describe equity claims:** distinguish the corporation, an exchange, a stock index, and an exchange-traded fund.
- **Compute growth rates:** calculate continuously compounded growth rates and convert them to log returns using  $r = (\Delta t)g$ .
- **Test return models:** use return distributions and autocorrelation to check whether a model matches observed market data.

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Companion notebook: [L3a lecture notebook](#)

# Lectures and Examples

## Lecture:

CHEME-5660-L3a-Lecture-Equity-Exchanges-Return-StylizedFacts-Fall-2026.ipynb

**Example:** CHEME-5660-L3a-SVD-SP500-Example-Fall-2026.ipynb

Use SVD to identify common patterns in the growth rates of S&P 500 constituents.

**Example:** CHEME-5660-L3a-StylizedFacts-Example-Fall-2026.ipynb

Compare distribution fits, inspect tail-index stability, and contrast the memory of signed growth with the memory of growth magnitude.

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**Link:** <https://github.com/varnerlab/CHEME-5660-CourseRepository-Fall-2026.git>

# Exchanges, Stocks, and ETFs

Before measuring an equity return, separate the marketplace from the securities traded there. These objects interact, but they are not interchangeable.

- An exchange is an organized venue that matches orders to buy and sell securities.
- An order is the instruction a broker submits on an investor's behalf; order types differ in how they set the price.
- A stock is a residual ownership claim on a corporation.
- An exchange-traded fund, or ETF, is a security whose shares represent an interest in a portfolio and trade during the day.

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**Today:** a market price is the price available at a particular time and trading venue.

## Exchanges Connect Buyers and Sellers

An exchange brings orders together and supports **price discovery**, the process by which trading establishes a market price, and **liquidity**, the ability to trade without moving that price excessively.

- Exchanges impose listing standards and trading rules; after execution, separate clearing and settlement infrastructure completes the transfer of securities and funds.
- U.S. equities trade across more than a dozen competing exchanges; the three largest are NYSE, NASDAQ, and Cboe BZX.
- A broker may route one order to a venue or divide it among venues, subject to available prices, liquidity, and the order's instructions.

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**Source:** [SEC list of registered national securities exchanges](#); the list also includes options-only venues.

# Orders Are Instructions to Trade

An order names the security, the direction (buy or sell), the quantity, and how the price may be set. The common order types differ in how they trade off immediacy against price control.

- **Market order:** executes as soon as possible at the best price currently available. Execution is nearly certain; the price is not.
- **Limit order:** executes only at a stated limit price or better: at or below it for a buy, at or above it for a sell. The price is controlled; execution is not guaranteed.
- **Stop and stop-limit orders:** wait until the price reaches a trigger level, then become a market order or a limit order.

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More order types: [Investor.gov, Types of Orders](https://www.investor.gov/types-of-orders).

# From Order Books to the NBBO

Limit orders that do not execute immediately rest on the venue's **order book**, the collection of outstanding buy and sell limit orders for a security.

- The highest resting buy price is the venue's displayed **bid**, the lowest resting sell price is its displayed **ask**, and the difference is the bid-ask spread.
- An arriving market order to buy trades at the ask; one to sell trades at the bid.
- Each venue keeps its own book, so each venue displays its own bid and ask. The **National Best Bid and Offer**, or NBBO, reports the highest displayed bid and the lowest displayed ask across the national market system.

A broker must fill a customer order at the national best price or better: a buyer pays no more than the NBBO ask, and a seller receives no less than the NBBO bid.



# Order Books and the NBBO for Stock XYZ

## Order books and the NBBO

National Best Bid and Offer

NASDAQ • XYZ

| SELL / ASK | SIZE | PRICE |
|------------|------|-------|
|------------|------|-------|

|  |     |       |
|--|-----|-------|
|  | 200 | 50.07 |
|--|-----|-------|

|  |     |       |
|--|-----|-------|
|  | 400 | 50.05 |
|--|-----|-------|

|      |     |       |
|------|-----|-------|
| BEST | 100 | 50.04 |
|------|-----|-------|

— spread = \$0.03 —

BUY / BID

|      |     |       |
|------|-----|-------|
| BEST | 300 | 50.01 |
|------|-----|-------|

|  |     |       |
|--|-----|-------|
|  | 500 | 50.00 |
|--|-----|-------|

|  |     |       |
|--|-----|-------|
|  | 200 | 49.99 |
|--|-----|-------|

TOP OF BOOK • XYZ

| VENUE | BID | ASK | SPREAD |
|-------|-----|-----|--------|
|-------|-----|-----|--------|

|      |       |       |      |
|------|-------|-------|------|
| NYSE | 50.00 | 50.05 | 0.05 |
|------|-------|-------|------|

|        |       |       |      |
|--------|-------|-------|------|
| Nasdaq | 50.01 | 50.04 | 0.03 |
|--------|-------|-------|------|

|          |       |       |      |
|----------|-------|-------|------|
| Cboe BZX | 50.02 | 50.06 | 0.04 |
|----------|-------|-------|------|

|     |       |       |      |
|-----|-------|-------|------|
| IEX | 50.01 | 50.06 | 0.05 |
|-----|-------|-------|------|

best bid and best ask sit  
on different venues

NBBO • XYZ

BEST BID

**\$50.02**

BEST ASK

**\$50.04**

Cboe BZX

Nasdaq

**spread \$0.02**

tighter than any single  
venue's own spread

# Stocks Are Claims on Firms

A stock, also called a share or **equity security**, represents fractional ownership in a corporation. It is a residual claim, not a promise of fixed repayment.

## What the shareholder may receive

- Voting rights and cash distributions such as dividends.
- A claim on remaining assets only after contractual creditors are paid.
- A capital gain if another investor later pays a higher share price.

Limited liability ordinarily caps the shareholder's loss at the invested amount. It does not protect the share price from falling to zero.

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**Source:** [U.S. Securities and Exchange Commission, Stocks](#)

# ETFs Are Claims on Portfolios

An ETF pools assets into a portfolio and issues shares that trade on an exchange throughout the day. Each share represents an interest in that portfolio.

- A broad-market fund such as SPY, which tracks the Standard & Poor's 500 index, can provide diversified exposure; sector, leveraged, inverse, and single-stock ETFs may be highly concentrated.
- The **net asset value**, or NAV, is the fund's assets minus its liabilities, divided by shares outstanding.
- An ETF has both a market price and a NAV per share. They are usually close, but the ETF can trade at a premium or discount to NAV.

Like a stock, an ETF produces an observable sequence of market prices.

# Reward

We measure price reward with the **continuously compounded growth rate**, denoted by  $g$ . Let  $S_{j-1}^{(i)} > 0$  and  $S_j^{(i)} > 0$  be consecutive prices for firm  $i$ , separated by  $\Delta t_j > 0$  years. The price evolution is:

$$S_j^{(i)} = S_{j-1}^{(i)} \exp(g_{j,j-1}^{(i)} \Delta t_j) .$$

Solving for the rate gives:

$$g_{j,j-1}^{(i)} = \underbrace{\frac{1}{\Delta t_j} \log \left( \frac{S_j^{(i)}}{S_{j-1}^{(i)}} \right)}_{\text{growth rate}} .$$

The growth rate has units of inverse years. Its sign records whether the price increased, decreased, or remained unchanged. The notebook computes  $g$  from closing-price series, so dividends and other cash distributions are not included.

# Growth Rates and Log Returns

Let  $g_t$  be the continuously compounded growth rate over an interval of length  $\Delta t_t > 0$  years. The corresponding **log return**  $r_t$  is dimensionless and is given by:

$$\underbrace{r_t = (\Delta t_t)g_t}_{\text{log return}} = \log\left(\frac{S_t}{S_{t-1}}\right).$$

Log returns add across adjacent intervals. For positive prices  $S_0, S_1, \dots, S_T$ , the cumulative log return is:

$$\sum_{t=1}^T r_t = \underbrace{\sum_{t=1}^T (\Delta t_t)g_t}_{\text{time-weighted growth}} = \log\left(\frac{S_T}{S_0}\right).$$

Use  $g$  to compare annualized rates. Convert to  $r$  before aggregating across time.

# Excess Growth Compares an Asset with a Benchmark

Let  $g_t$  be the equity growth rate and let  $g_{f,t}$  be a declared benchmark growth rate over the same interval of length  $\Delta t_t$  years. The **excess growth rate** and excess log return are given by:

$$\underbrace{\bar{g}_t = g_t - g_{f,t}}_{\text{excess growth}}, \quad \underbrace{\bar{r}_t = (\Delta t_t) \bar{g}_t}_{\text{excess log return}}.$$

- $\bar{g}_t > 0$ : the equity price grew faster than the benchmark.
- $\bar{g}_t < 0$ : the equity price grew more slowly than the benchmark.
- $\bar{g}_t = 0$ : both produced the same growth rate over the interval.

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**Careful:** the benchmark must use the same currency, horizon, and time units. A positive excess rate is not a guarantee of positive wealth.

# Algorithm

Let  $\mathcal{D}_i = \{S_1^{(i)}, \dots, S_T^{(i)}\}$  be a positive price series for firm  $i$ , let  $\Delta t > 0$  be the interval in years, and let  $g_f$  be the benchmark growth rate.

1. For  $t = 2, \dots, T$ , look up  $S_{t-1}^{(i)}$  and  $S_t^{(i)}$ .
2. Compute the excess growth observation:

$$\bar{g}_t^{(i)} \leftarrow \underbrace{\frac{1}{\Delta t} \log \left( \frac{S_t^{(i)}}{S_{t-1}^{(i)}} \right)}_{g_t^{(i)}} - g_f.$$

3. Store  $\bar{g}_t^{(i)}$ . The  $T$  prices produce  $T - 1$  observations.

Set  $g_f = 0$  when the required output is the growth rate rather than the excess growth rate.

# Common Patterns Across Many Assets

Let  $\mathbf{G} \in \mathbb{R}^{m \times n}$  collect centered growth rates for  $m$  dates and  $n$  assets. Its singular value decomposition factors  $\mathbf{G}$  into date patterns  $\mathbf{u}_j$  (the columns of  $\mathbf{U}$ ), asset patterns  $\mathbf{v}_j$  (the columns of  $\mathbf{V}$ ), and nonnegative weights  $\sigma_j$  (the diagonal of  $\Sigma$ ):

$$\mathbf{G} = \mathbf{U}\Sigma\mathbf{V}^\top = \sum_{j=1}^{\text{rank}(\mathbf{G})} \underbrace{\sigma_j \mathbf{u}_j \mathbf{v}_j^\top}_{\text{rank-one pattern}}, \quad \sigma_1 \geq \sigma_2 \geq \dots \geq 0.$$

Large singular values identify common patterns that account for much of the variation across assets.

SVD finds patterns in the data; it does not explain their economic causes. Results depend on centering, scaling, missing-data treatment, and the selected assets.

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**Notebook:** CHEME-5660-L3a-SVD-SP500-Example-Fall-2026.ipynb



# Risk

Risk is uncertainty about loss or gain. A common dispersion measure is the sample standard deviation of the growth rate, denoted by  $\sigma_g$ .

Suppose  $T$  prices produce the  $T - 1$  observations  $g_2, g_3, \dots, g_T$ . Let  $g'$  be their sample mean. The sample volatility is:

$$g' = \frac{1}{T-1} \sum_{t=2}^T g_t, \quad \sigma_g = \underbrace{\sqrt{\frac{1}{T-2} \sum_{t=2}^T (g_t - g')^2}}_{\text{growth-rate volatility}}.$$

The statistic has the same inverse-year units as  $g$ . It measures dispersion around the sample mean, not tail probability, drawdown, liquidity, or the magnitude of a specific loss.

# Other Risk Measures

Choose a risk measure that matches the type of loss relevant to the decision.

- **Value at Risk**: a loss threshold exceeded with a stated small probability over a stated horizon.
- **Expected Shortfall**: the average loss beyond that threshold.
- **Maximum drawdown**: the largest peak-to-trough decline in value.
- **Downside deviation**: dispersion measured only below a declared target.
- **Beta**: sensitivity to a selected market benchmark.
- **Tracking error**: volatility of the return difference from a benchmark.

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**Ask:** what type of loss matters, over what horizon, and will this statistic detect it?

# Stylized Facts of Equity Growth Rates

A **stylized fact** is a recurring empirical pattern observed across many assets or markets. It is not an exact law for every ticker and sample. It is a pattern that a useful model should be able to reproduce.

- **Heavy tails**: large movements occur more often than a fitted Gaussian model predicts.
- **Weak linear autocorrelation**: signed growth rates often have little serial linear dependence at ordinary liquid-market horizons.
- **Volatility clustering**: large movement magnitudes occur near other large movements, and calm observations near other calm observations.

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**Sources:** [Cont \(2001\)](#); [Mandelbrot \(1963\)](#)

# Heavy Tailed Return Distributions

A power-law tail decays more slowly than an exponential tail. Let  $X$  be a random variable, let  $\alpha > 0$  be its **tail index**, and let  $L(x)$  be slowly varying. A two-sided power-law tail satisfies:

$$\mathbb{P}(|X| > x) \sim \underbrace{L(x)x^{-\alpha}}_{\text{tail probability}} \quad \text{as } x \rightarrow \infty.$$

A smaller  $\alpha$  means slower decay and a heavier tail. The classical strict thresholds identify moments that fail to exist for a pure power-law tail:

- $\alpha < 1$ : the mean fails to exist.
- $\alpha < 2$ : the variance fails to exist.
- $\alpha < 4$ : the fourth moment fails to exist.

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**Careful:** at the boundary values 1, 2, and 4, moment existence depends on finer tail detail. Heavy-tailed does not always mean power-law.

# Hill's Estimator of the Tail Index

Hill's estimator applies to positive observations. For signed growth rates, analyze one tail or the magnitude series  $|g|$ . Let  $X_{(1)} \geq X_{(2)} \geq \dots \geq X_{(n)} > 0$  be descending order statistics, and let  $k < n$  select the upper-order observations. The estimate is:

$$\hat{\alpha}_k = \underbrace{\left[ \frac{1}{k} \sum_{i=1}^k \log \left( \frac{X_{(i)}}{X_{(k+1)}} \right) \right]^{-1}}_{\text{estimated tail index}}.$$

The cutoff  $k$  balances bias and variance. One arbitrary cutoff is not a diagnostic: compute  $\hat{\alpha}_k$  over a range of  $k$  values and look for a stable region.

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**Companion example:** the stylized-facts notebook uses a Hill stability plot and treats  $k = \lfloor \sqrt{n} \rfloor$  as a reproducible reference, not a uniquely correct cutoff.

# Autocorrelation

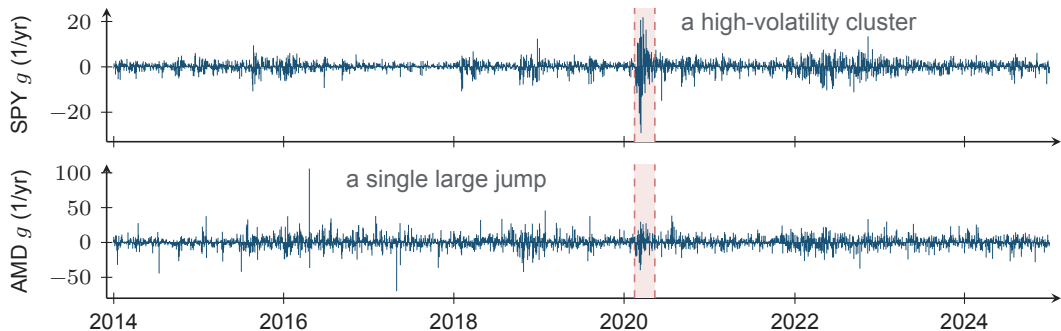
Let  $g_2, \dots, g_T$  be growth-rate observations with sample mean  $g'$ , and let  $k$  be a lag measured in sampling intervals. Define the empirical autocovariance and autocorrelation by:

$$\widehat{R}(k) = \frac{1}{T-1} \sum_{t=2}^{T-k} (g_t - g')(g_{t+k} - g'), \quad \underbrace{\rho(k) = \frac{\widehat{R}(k)}{\widehat{R}(0)}}_{\text{autocorrelation}}.$$

This biased normalization divides by the full sample size at every lag, so  $\rho(0) = 1$  and  $|\rho(k)| \leq 1$ . For many equity samples,  $\rho(k)$  is near zero at nonzero lags: past signed growth provides little **linear** help in predicting future signed growth.

Near-zero autocorrelation does **not** establish independence. Dependence can remain in growth magnitude, trading microstructure, or other nonlinear functions of the series.

# Volatility Clustering



**Volatility clustering:** large-magnitude days arrive in bursts of both signs, and the marked weeks are turbulent in both series; one large jump is not a cluster.

**Data:** daily SPY and AMD growth rates, 2014–2024, from the stylized-facts example dataset.

# Diagnosing Volatility Clustering

Let  $g_2^{(i)}, \dots, g_T^{(i)}$  be the growth-rate series for firm  $i$ , and let  $\overline{|g|}_i$  be the mean growth-rate magnitude. Magnitudes weight extreme days less heavily than squares and give a cleaner estimate. The empirical autocorrelation of growth-rate magnitudes is:

$$\hat{\rho}_i(k) = \frac{\sum_{t=2}^{T-k} \left[ |g_t^{(i)}| - \overline{|g|}_i \right] \left[ |g_{t+k}^{(i)}| - \overline{|g|}_i \right]}{\sum_{t=2}^T \left[ |g_t^{(i)}| - \overline{|g|}_i \right]^2}.$$

Volatility clustering appears when  $\hat{\rho}_i(k) > 0$  across many lags and fades slowly toward zero. A model can therefore match weak signed-growth ACF and still fail because it assumes fixed, independent volatility.

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**Why it matters:** conditional volatility changes risk forecasts, option values, hedge ratios, and capital allocation.



# Example

## Diagnosing Stylized Facts in Daily Equity Growth Rates

The companion notebook turns each stylized fact into a model check:

- Compare empirical growth rates with fitted Normal, Laplace, and Student- $t$  densities.
- Inspect a Hill stability plot instead of reporting one cutoff.
- Compare the ACF of signed growth with the ACF of growth magnitude.
- Translate the diagnostics into requirements for the next model.

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**Notebook:** CHEME-5660-L3a-StylizedFacts-Example-Fall-2026.ipynb

# Optional Advanced Material

**Advanced 1:** `advanced/order_book/`

CHEME-5660-L3a-Advanced-LimitOrderBook-Mechanics-Fall-2026.ipynb

**Advanced 2:** `advanced/market_impact/`

CHEME-5660-L3a-Advanced-Market-Impact-Models-Fall-2026.ipynb

**Advanced 3:** `advanced/stylized_facts/`

CHEME-5660-L3a-Advanced-StylizedFacts-Fall-2026.ipynb

Optional, not prerequisites for L3b. The first develops the mechanics of a price-time-priority limit-order book; the second distinguishes mechanical execution cost from linear, order-flow-imbalance, and transient-impact models; the third deepens the stylized-facts diagnostics and hands them off to L6.

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**Notebooks:** [lectures/week-3/L3a/advanced](#), also linked from the lecture notebook.

# Summary

This lecture introduced equity securities, continuously compounded growth rates, and recurring empirical patterns in financial return data.

## Key takeaways

- **Market infrastructure and securities are different.** Exchanges discover prices; stocks and ETFs are claims traded there.
- **Return statistics require conventions.** The growth rate annualizes and the log return aggregates across intervals; both are computed here from closing prices, without cash distributions.
- **Stylized facts are model requirements.** Heavy tails and clustered volatility expose failures of a fixed-variance independent model.

These diagnostics become tests for the models that follow.

That's a wrap. What are some of the interesting things we discussed today?